

Alexander Hasenbein

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EDUCATION

The Pennsylvania State University

B.S. Computer Science

B.S. Mathematics

University Park, PA

Expected May 2028

GPA: 3.5

WORK EXPERIENCE

Flex

Austin, TX

Software Engineering Intern

May 2026 - Aug 2026

- Engineered features from SQL Server views and optimized complex queries across production datasets of **4M+ rows**, implementing parallelized chunking to eliminate tempdb exhaustion.
- Designed web-based dashboards in C# (ASP.NET) with dynamic charts, tables, and filters, delivering real-time analytics for the process engineering team. Owned delivery of 2 HQ-wide projects, managing version control and roadmap tracking.
- Built internal data analysis tooling in Python (Plotly, pandas) to streamline reporting workflows, **reducing analysis time by 10%** against baseline (timed measurement against standard tools).

The Pennsylvania State University

Erie, PA

Undergraduate Researcher, Honors Option

Jan 2025 - May 2025

- Built a configurable Python benchmarking framework with pluggable strategy interfaces to compare optimization algorithms (Optuna-based Bayesian search, random search, multi-objective Pareto search) against PyTorch models, quantifying accuracy-efficiency tradeoffs.
- Engineered a data pipeline and full evaluation suite (MAE, RMSE, R^2 , per-class error, inference speed) with interactive dashboards for trial exploration, ensuring reproducible, interpretable results.

PROJECTS

DevCord – Multi Agent AI Orchestration Platform | 1st Place - AI Agents & LLMs: Behrend Hackathon

<https://github.com/AHasenbein/DevCord-Multi-Agent>

TypeScript, Discord.js, OpenRouter

- Architected specialized agents (Manager, Dev, QA, Command) with scoped roles and task handoff in strict TypeScript using Discord.js and OpenRouter, with modular model routing to swap LLMs per agent without modifying orchestration logic.
- Implemented autonomous project kickoff, thread-based task decomposition, persistent run/state tracking with pause/resume/cancel controls, plus voice-first Q&A via local Whisper STT/TTS in Discord voice channels.
- Leveraged Claude Code as an AI pair-programmer for rapid prototyping, debugging, and test coverage, configuring rules enforcing TypeScript strict mode and modular architecture to keep AI-suggested changes swappable across agent models.

Particles Interlinked

Next.js, React, Three.js, Node.js

- Developed a complex real-time particle simulation system using Three.js with WebGL rendering, applying physics concepts such as particle interactions, gravitational attraction, and collision detection implemented efficiently in JavaScript.
- Integrated performance monitoring and leveraged WebGL for hardware-accelerated graphics rendering, demonstrating strong skills in performance tuning and UI/UX for interactive simulations, **increasing fps by ~22% consistently**.
- Migrated from a 2D to a 3D library, improving visual dynamism and **rendering speed** over the prior implementation.

BowieForum

React, MongoDB, SQL, Node.js

- Designed and deployed full-stack student forum (React, Node.js, MongoDB) serving 150+ active users.
- Optimized SQL queries and indexing strategy, reducing average query time and server load under concurrent traffic.
- Designed responsive frontend and RESTful backend architecture. Created a responsive UI for better UX.

SKILLS & ACCOMPLISHMENTS

- Programming Languages:** Java, C++, C#, JavaScript, TypeScript, Python, SQL, VB.NET, HTML, CSS, Go
- Frameworks:** React, Next.js, Express.js, Spring, ASP.NET, FastAPI, Flask, pandas
- Tools & Technologies:** Git, Docker, MongoDB, SQL Servers, Node.js, OpenRouter, CI/CD Pipelines, AWS, Azure
- Accomplishments:** Commonwealth Engineering Scholar, Behrend Honors Program, Nittany AI Leadership Academy, ELP